

Unsupervised is the most interesting as it relies on the AI to come up with its own solution. Traditional methods of machine learning require you to introduce rule or algorithms, or require the data to be pre-labeled, but by doing its own labeling and feature finding it can save the programmer some time while also possibly identifying useable features that people might not have thought about.

Identifying the data that will be used and making sure you separate and process the data so that only what you need is processed by the model.

Applying this to my field of interest (cybersecurity) I would need to gather the right kind of data useful to my application. Ideally, gather logs of user activity, emails that were flagged, or maybe files containing structures that I want protected. Then I would have to choose how to train a model. Depending on if I know what I am looking for, I can choose to create a model that uses labeled example to save time, or if I suspect that a threat is incoming in a form that I don't expect, I could choose to employ an unsupervised model where it creates its own patterns and then simply looks for deviations from those patterns.

Most of my questions are code related, as well as if there is any proper way to introduce certain elements into the code or if there is some leniency in the order. Additionally, one question more related to this module would relate to how to implement reinforcement learning. Is it applicable after employing supervised or unsupervised learning, and if so is there a reason why we can't just always apply reinforcement learning everywhere to strengthen models.